

■ NuclearCraft: Plutonium Age

Plugin Guide & Crafting Reference
Version 1.0.1 · PaperMC 1.21+ · Java 21

A comprehensive guide covering all items, recipes, mechanics, and systems added by the NuclearCraft: Plutonium Age plugin. Radiation is not just a hazard — it is a resource, a weapon, and a way of life.

■ **Radiation permeates every system in this plugin. Handle all plutonium materials with care.**

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1. Custom Items & Materials

All NuclearCraft items are crafted using the **Custom Model Data (CMD)** system — they visually replace vanilla items but retain full custom behaviour via the resource pack.

Core Materials

Item	Base Item	CMD	Description
Radioactive Core	Magma Cream	1101	Pulsing nuclear energy core. Crafting material & forge fuel.
Raw Plutonium Fragment	Prismarine Crystals	1102	Mined from Plutonium Ore. Must be refined before use.
Refined Plutonium Ingot	Echo Shard	1103	Output of the Nuclear Smelter. Used in all gear crafting.
Irradiated Heart	Heart of the Sea	1106	Dropped by Irradiated Zombies. Boss summon component.
Titan Core	Nether Star	1107	Crystallised essence of the Plutonium Titan. Boss summon.
Industrial Fabric	Yellow Wool	1315	Radiation-hardened material for Hazmat Suit crafting/repair.

Consumables & Cure Items

Item	Base Item	CMD	Effect
Healing Petal	Pink Petals	1105	Harvested from Mutated Healing Plants. Cure ingredient.
Radiation Antidote	Honey Bottle	1301	Right-click to consume. Clears all radiation & infection. No immunity.
Radiation Serum	Glass Bottle	1302	Right-click to consume. Clears radiation + 10 min immunity. Rare.

Machines

Item	Base Item	CMD	Function
Nuclear Smelter	Blast Furnace	1201	Refines Raw Plutonium Fragments into Refined Ingots. Requires fuel.
Nuclear Forge	Smithing Table	1401	Upgrades Plutonium & Hazmat gear from MK-I to MK-IV. Requires Radioactive Cores.

Tools & Weapons (Quick Reference)

Item	Base Item	CMD	Notes
Radiation Drill	Diamond Pickaxe	1108	Only tool that safely mines Plutonium Ore without radiation burst.
Plutonium Sword	Netherite Sword	1301	High damage + radiation-on-hit. Gains mastery XP with each kill.
Plutonium Axe	Netherite Axe	1302	Cleave damage + axe-class mastery bonuses.
Plutonium Pickaxe	Netherite Pickaxe	1303	Fastest mining speed tier.
Plutonium Shovel	Netherite Shovel	1304	Converts terrain to radioactive farmland on use.
Plutonium Hoe	Netherite Hoe	1305	Tills radioactive farmland. Required for mutated crop planting.
Plutonium Arrow	Arrow	1314	Fired from any bow. Injects radiation directly into target. Phase 9 PvP weapon.

2. Radiation System

Radiation is the central mechanic of NuclearCraft. It accumulates from exposure to plutonium ore, radioactive items, irradiated zones, mobs, and PvP combat. As radiation increases, the player progresses through four stages — each with escalating debuffs.

Radiation Stages

Stage	Threshold	Debuffs	Visual Effect
0 — Clean	0	None	Normal HUD
1 — Irradiated	1–499	Nausea, mild slowness	Green tint, subtle HUD glow
2 — Contaminated	500–999	Poison, slowness II, weakness	Screen distortion, green particles
3 — Critical	1000+	Wither, blindness, severe slowness, hunger	Heavy screen effects, constant damage

Radiation Sources

Source	Radiation Gain	Notes
Standing near Plutonium Ore	Configurable per tick	Range-based; blocked by full hazmat suit
Mining Plutonium Ore (no drill)	Large burst	Use Radiation Drill to mine safely
Irradiated Zombie proximity	Moderate per second	Alpha Zombies emit more than standard
Radiation Cloud	High while inside	Created by Alpha Zombie deaths
Radiation Night event	Passive area-wide	Triggered at configurable intervals
Plutonium Arrow hit	Injected directly	PvP — Phase 9
Radiation Aura (PvP)	Per aura tick	From MK-III/IV equipment — Phase 9
Radioactive Core (held)	Minor passive	Configurable

Reducing Radiation

- Radiation decays naturally over time (configurable rate).
- **Radiation Antidote** — instantly clears all radiation.
- **Radiation Serum** — clears radiation and grants 10 minutes of immunity.
- **Hazmat Suit** — reduces radiation gain by up to 60% when wearing all 4 pieces.
- **Plutonium Armor MK-III+** — built-in radiation resistance that scales with upgrade tier.

3. Nuclear Smelter

The Nuclear Smelter (CMD 1201, base: Blast Furnace) is an early-game machine that converts Raw Plutonium Fragments into Refined Plutonium Ingots. It is required to progress into gear crafting.

Obtaining

Give command: `/nc give <player> nuclear-smelter`

Crafting: Configured in `smelter.yml` (default recipe uses Blast Furnace + Radioactive Cores).

Operation

- Right-click the Nuclear Smelter item in the world (it functions as a placed block alternative).
- Add **Raw Plutonium Fragments** to the input slot.
- Add fuel: **Coal**, **Blaze Rod**, or **Lava Bucket**.
- Each fragment smelts into one **Refined Plutonium Ingot**.
- Smelting speed and fuel efficiency are configurable.

Fuel	Burn Time (ticks)	Ingots per Fuel
Coal	1600	~8
Blaze Rod	2400	~12
Lava Bucket	20000	~100 (bucket returned)

4. Irradiated Zombies & Mob Events

NuclearCraft adds two zombie variants and two server-wide mob events.

Zombie Types

Type	Health	Damage	Radiation Aura	Special Drops	Spawn Condition
Irradiated Zombie	30 HP	+4 atk	Moderate	Irradiated Heart (chance)	Natural + /nc zombie spawn irradiated
Alpha Zombie	60 HP	+8 atk	High	Radioactive Core, Irradiated Heart	Low chance near ore, /nc zombie spawn alpha

Events

Event	Trigger	Effect	Duration
Radiation Night	Configurable interval (default nightly)	Passive radiation gain for all players on surface	Full night cycle
Zombie Surge	Configurable interval or /nc zombie surge start	Mass irradiated zombie spawns near players	Configurable ticks
Radiation Cloud	Alpha Zombie death	Lingering radiation cloud at death location	Configurable ticks

5. Plutonium Ore & Mining

Plutonium Ore is the foundation of all progression. It generates underground and emits a radiation field that damages nearby unprotected players.

Property	Value
Block type	Custom ore (configurable stone host)
Generation depth	Y -64 to Y 16 (configurable)
Vein size	1–4 blocks (configurable)
Radiation radius	5 blocks (configurable)
Safe mining tool	Radiation Drill only — any other tool triggers radiation burst
Drop	Raw Plutonium Fragment (1–3, configurable Fortune bonuses)
Fortune interaction	Fortune I = +1, Fortune II = +2, Fortune III = +3 max

6. Equipment — Tools, Weapons & Armor

All equipment is crafted from **Refined Plutonium Ingots** and **Industrial Fabric** (for armor). Equipment can be upgraded via the Nuclear Forge (see Section 8).

Hazmat Suit (Radiation Protection)

Piece	CMD	Base Item	Armor	Radiation Reduction
Hazmat Helmet	1306	Leather Helmet	3	15% per piece
Hazmat Chestplate	1307	Leather Chestplate	8	15% per piece
Hazmat Leggings	1308	Leather Leggings	6	15% per piece
Hazmat Boots	1309	Leather Boots	3	15% per piece
Full set bonus				Total: 60% radiation reduction (max configurable)

Plutonium Armor (Combat + Resistance)

Piece	CMD	Base Item	Armor	Special
Plutonium Helmet	1310	Netherite Helmet	3	Radiation resistance scales with MK tier
Plutonium Chestplate	1311	Netherite Chestplate	8	Radiation resistance scales with MK tier
Plutonium Leggings	1312	Netherite Leggings	6	Radiation resistance scales with MK tier
Plutonium Boots	1313	Netherite Boots	3	Radiation resistance scales with MK tier
Full set bonus				Radiation Aura at MK-III/IV (damages nearby players in PvP)

Weapon Stats (Base / Before Upgrades)

Weapon	CMD	Damage	Attack Speed	Special
Plutonium Sword	1301	+7 atk dmg	1.6/s	Radiation-on-hit, combo system, sword mastery
Plutonium Axe	1302	+9 atk dmg	0.9/s	Radiation-on-hit, axe mastery, critical surge
Plutonium Arrow	1314	Bow base	—	Injects radiation on hit; bow mastery XP on kills
Radiation Drill	1108	Diamond equiv	—	Safe plutonium ore mining; no mining radiation burst

7. Farming — Mutated Crops & Cure Crafting

NuclearCraft adds a radiation-based farming system. Mutated Seeds grow only on Radioactive Farmland (created by right-clicking dirt with a Plutonium Shovel) and produce Healing Petals — the core ingredient for radiation cure items.

Farming Steps

1. Obtain **Mutated Seeds** — dropped by Irradiated Zombies or given via `/nc farming give mutated-seed`.
2. Create **Radioactive Farmland** — right-click any dirt/grass block with the **Plutonium Shovel**.
3. Plant the seed on farmland using the **Plutonium Hoe**.
4. Water as normal. Crops grow faster in higher radiation zones.
5. Harvest to obtain **Healing Petals**.

Cure Crafting Recipes

Item	Ingredients	Crafting Type	Effect
Radiation Antidote	3× Healing Petal + 1× Honey Bottle + 1× Gold Ingot	Crafting Table (3×3)	Clears all radiation & debuffs
Radiation Serum	2× Radiation Antidote + 1× Refined Plutonium Ingot + 1× Radioactive Core	Crafting Table (3×3)	Clears radiation + 10 min immunity

Admin commands: `/nc farming give <player> mutated-seed|healing-petal|radiation-antidote|radiation-serum [amount]`

`/nc farming growall` – instantly grows all mutated crops on the server.

8. Nuclear Forge & Equipment Upgrades

The Nuclear Forge (CMD 1401) upgrades Plutonium and Hazmat equipment from **MK-I** through **MK-IV**. Each tier increases damage, armor, radiation resistance, and unlocks new passive abilities such as the Radiation Aura.

Upgrade Tiers

Tier	Radioactive Cores Required	Damage Bonus	Armor Bonus	Special
MK-I	Base (no forge needed)	+0	+0	Base equipment
MK-II	5 Radioactive Cores	+15%	+5%	Slight radiation resistance bonus
MK-III	15 Radioactive Cores	+30%	+10%	Radiation resistance +25%; Radiation Aura unlocked for armor
MK-IV	40 Radioactive Cores	+50%	+20%	Maximum resistance; Radiation Aura at full strength; set bonus

Radiation Aura (MK-III/IV Armor)

When wearing MK-III or MK-IV Plutonium Armor, the player emits a **Radiation Aura** that passively irradiates nearby players in PvP. Aura range and damage scale with tier. The Aura replaces the old passive aura system in Phase 9 (AdvancedAuraManager).

Forge Commands

```
/nc forge give <player> nuclear-forge – give a Nuclear Forge.  
/nc forge energy set|add|clear <player> [amount] – manage forge energy.  
/nc forge upgrade mk2|mk3|mk4 <player> – force-upgrade held item.  
/nc forge stats [player] – view upgrade statistics.
```

9. Advanced Combat & PvP Radiation Mechanics

Phase 9 transforms radiation from a passive hazard into an active combat weapon. Hitting players with Plutonium weapons injects radiation. Combo chains amplify radiation bursts. Weapon mastery improves damage over time. The Radiation Aura damages nearby enemies passively.

Core Combat Systems

System	Description
Radiation Injection	Plutonium Sword/Axe hits inject radiation into the victim. Amount scales with attacker's weapon tier.
Combo System	Landing consecutive hits builds a combo. Higher combos amplify radiation injection (configurable multiplier).
Critical Surge	After a combo threshold, a radiation surge fires — dealing bonus radiation to all nearby enemies.
Weapon Mastery	Each weapon type (sword, axe, bow) has its own XP track. Higher mastery = higher base radiation injection.
Plutonium Arrow	Fires from any bow. Injects radiation on impact. Bow mastery XP gained on kills.
Radiation Aura (PvP)	MK-III/IV armor emits aura that irradiates nearby players continuously. Managed by AdvancedAuraManager.
Infection System	High-radiation hits can apply a lingering infection debuff that causes radiation to spread/linger.

PvP Radiation Kill Types

Kill Type	Tracked Stat	Advancement
Standard PvP kill	pvpKills	—
Radiation kill	pvpRadiationKills	RADIOACTIVE_WARRIOR (first), RADIATION_MASTER (10,000 total rad inflicted)
Plutonium Arrow kill	pvpArrowKills	NUCLEAR_ARCHER
Aura kill	pvpAuraKills	—
Contamination (stage↑)	—	CONTAMINATOR

Mastery Levels

Mastery XP	Level	Bonus
0	Novice	+0% radiation injection
500	Adept	+10% radiation injection
1500	Expert	+25% radiation injection
3000	Master	+50% radiation injection
6000	Grandmaster	+80% radiation injection + Surge damage boost

Combat Commands

```
/nc combat stats [player] — view PvP kill statistics.  
/nc combat mastery [player] — view weapon mastery levels.  
/nc combat radiation [player] — view radiation combat data (arrow hits, aura damage).
```

10. Commands Reference

All commands use the `/nuclearcraft` or `/nc` alias.

General

Command	Permission	Description
<code>/nc help</code>	—	Show help menu
<code>/nc version</code>	—	Display plugin version
<code>/nc info</code>	—	Show runtime info
<code>/nc reload</code>	<code>nuclearcraft.reload</code>	Reload all configuration files
<code>/nc debug</code>	<code>nuclearcraft.debug</code>	Toggle debug mode
<code>/nc radiation check</code>	—	View your own radiation status

Admin Subcommands

Subcommand	Key Arguments	Description
<code>/nc give</code>	<code><player> <item-id> [amount]</code>	Give any custom item
<code>/nc radiation</code>	<code>add remove set clear <player> [amount]</code>	Manage player radiation
<code>/nc zombie</code>	<code>spawn irradiated alpha <player></code>	Spawn zombie near player
<code>/nc zombie surge</code>	<code>start stop</code>	Toggle zombie surge event
<code>/nc ore spawn</code>	<code><player></code>	Spawn plutonium ore near player
<code>/nc ore give</code>	<code><player> fragment drill [amount]</code>	Give ore-related items
<code>/nc smelter give</code>	<code><player> [amount]</code>	Give a Nuclear Smelter
<code>/nc equipment give</code>	<code><player> <slot> [mk]</code>	Give equipment piece
<code>/nc farming give</code>	<code><player> <item> [amount]</code>	Give farming items
<code>/nc farming growall</code>	—	Instantly grow all crops
<code>/nc forge give</code>	<code><player> [amount]</code>	Give a Nuclear Forge
<code>/nc forge energy</code>	<code>set add clear <player> [amount]</code>	Manage forge energy
<code>/nc forge upgrade</code>	<code>mk2 mk3 mk4 <player></code>	Force upgrade held item
<code>/nc combat stats</code>	<code>[player]</code>	View PvP kill stats
<code>/nc combat mastery</code>	<code>[player]</code>	View weapon mastery

11. Permissions

Permission Node	Default	Description
nuclearcraft.admin	op	Parent node — grants all admin permissions
nuclearcraft.admin.radiation	op	Manage player radiation
nuclearcraft.admin.zombie	op	Spawn zombies, trigger surge
nuclearcraft.admin.ore	op	Spawn ore, give ore items
nuclearcraft.admin.smelter	op	Give smelters, debug smelter
nuclearcraft.admin.equipment	op	Give equipment, manage gear
nuclearcraft.admin.farming	op	Give seeds/cures, growall
nuclearcraft.admin.forge	op	Give forge, manage upgrades
nuclearcraft.admin.combat	op	View combat stats, mastery data
nuclearcraft.give	op	Use /nc give command
nuclearcraft.reload	op	Use /nc reload command
nuclearcraft.debug	op	Use /nc debug command

12. Configuration Files

All configuration files are located in `plugins/NuclearCraft/`. Use `/nc reload` to apply changes without restarting the server.

File	Controls
<code>config.yml</code>	General settings: language, debug mode, global enable flags
<code>radiation.yml</code>	Radiation gains, decay rate, stage thresholds, armor reduction, progression intervals
<code>zombies.yml</code>	Zombie health/damage/aura, spawn weights, drop tables
<code>boss.yml</code>	Plutonium Titan stats, arena settings, spawn conditions
<code>ore.yml</code>	Ore generation: depth range, vein size, radiation radius, drop amounts
<code>smelter.yml</code>	Fuel burn times, smelt speed, recipe
<code>toolstats.yml</code>	Tool/weapon damage, attack speed, durability, mining speed
<code>armors.yml</code>	Armor values, radiation resistance per piece, set bonuses
<code>equipment.yml</code>	Equipment upgrade multipliers, aura range/damage, repair costs
<code>forge.yml</code>	Forge energy capacity, core cost per tier, overload mechanics, upgrade recipes
<code>farming.yml</code>	Crop growth speed, farmland creation, harvest yield, cure recipes
<code>combat.yml</code>	PvP radiation injection amounts, combo multipliers, mastery XP thresholds, surge config, aura tick rate
<code>messages.yml</code>	All player-facing messages (supports MiniMessage formatting)

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■ All radiation thresholds, damage values, and drop rates are fully configurable.